

Discontinuous Galerkin Methods for Timoshenko Beams

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Summary. We devise a family of discontinuous Galerkin methods for the Timoshenko beam problem. Sufficient conditions for the existence and uniqueness of the approximation are given. The method allows arbitrary meshes and arbitrary polynomial degrees within the mesh, and hence is suitable for hp adaptivity. Numerical results showing optimal and exponential convergence are provided. These features of the method render it appealing for other problems in structure mechanics such as, plates, shells etc.

1 Introduction

In this paper, we introduce and numerically study discontinuous Galerkin (DG) for Timoshenko beams. The Timoshenko beam model, see [1] and [2], can be written as

$$\frac{dw}{dx} = \theta(x) - \frac{T(x)}{(GA)(x)}, \quad \frac{d\theta}{dx} = \frac{M(x)}{(EI)(x)}, \quad \frac{dM}{dx} = T(x), \quad \frac{dT}{dx} = q(x), \quad (1)$$

where $x \in \Omega = (0, L)$. Here, the unknowns are the transverse displacement w , the rotation of the transverse cross-section of the beam θ , the bending moment M , and the shear force T . The material and geometrical properties of the beam are characterized by the shear modulus G , the cross-section area A , the Young modulus E , and the moment of inertia I . The transverse load, q , is part of data of the problem. To complete the model and ensure the existence and uniqueness of its solution, we must impose suitable boundary conditions; we take, for example,

$$w(0) = w_0 \quad M(0) = M_0 \quad w(L) = w_L \quad M(L) = M_L. \quad (2)$$

Our long-term goal is to investigate the possible advantages of DG methods in computational structural mechanics. In this paper, we begin our efforts by

studying how to properly devise DG methods for the Timoshenko beam. In a forthcoming paper, we give a complete error analysis of these methods and show that they can easily overcome the so-called shear locking.

In [4] Arnold analyzed the continuous version of the Galerkin method. He proved error estimates which degenerate as the aspect ratio of the beam tends to zero, and hence the method suffers from shear locking. In the same paper he proved that locking is overcome if we use the so-called reduced integration technique. These findings are verified by numerical experiments. For the relationship between mixed finite element methods and reduced integration techniques we refer to [5]. In [6], Li analyzed the p and hp versions of the same method and proved error estimates independent of the aspect ratio of the beam. This is consistent with the well known fact that locking can be overcome by increasing the polynomial degree of the approximations. Our preliminary results indicate that the DG methods overcome locking even if we approximate all the unknowns with piecewise constants and do not use reduced integration. This and other features render DG appealing for other problems in structure mechanics, such as plates, shells and elasticity. Of course, there are additional issues involved in these problems; Reissner-Mindlin plates have boundary layers, elasticity problems have volumetric locking and shells exhibit membrane locking. For a recent DG method for the Reissner-Mindlin plate see [7]. Arnold and Falk provided a theoretical analysis of the boundary layers in [8], and a uniformly accurate finite element method in [9]. For a locking-free finite element method for shells see [10].

The paper is organized as follows. In section 2, we introduce the weak formulation we are going to use to define the DG methods. Then, in section 3, we introduce their general form. In section 4, we introduce what we call a discrete energy identity which we use, in section 5, to establish conditions that ensure existence and uniqueness of the approximate solution. In section 6, we present numerical results showing that the method can achieve optimal h -convergence as well as exponential convergence. We end in section 7 with some concluding remarks.

2 The weak formulation for the continuous case

To display the weak formulation we use to define the DG methods, we need to introduce some notations. Let $\mathcal{T} = \{I_j = (x_{j-1}, x_j), j = 1, \dots, N\}$ be a triangulation of the computational domain $\Omega = (0, L)$; we assume that the nodes x_i are such that $0 = x_0 < x_1 < \dots < x_{N-1} < x_N = L$. Then, we write

$$(\varphi, \psi) := \sum_{j=1}^N \int_{I_j} \varphi \psi \quad \text{and} \quad \langle R, \llbracket \varphi n \rrbracket \rangle := \sum_{j=0}^N R(x_j) \llbracket \varphi n \rrbracket(x_j).$$

Here, R is a function defined on the set of nodes $\mathcal{E}_h := \{x_0, x_1, \dots, x_N\}$. The *jump* of the function φ , $\llbracket \varphi n \rrbracket$, is defined as follows. If the node e is in

$\mathcal{E}_h^o := \{x_1, x_2, \dots, x_{N-1}\}$, then we take $\llbracket \varphi n \rrbracket(e) = \varphi(e^+)n_e^+ + \varphi(e^-)n_e^-$, where $\varphi(e^\pm) := \lim_{\epsilon \downarrow 0} \varphi(e - \epsilon n_e^\pm)$ and $n_e^\pm = \mp 1$. For the boundary nodes, we take $\llbracket \varphi n \rrbracket(0) = -\varphi(0^+)$, $\llbracket \varphi n \rrbracket(L) = \varphi(L^-)$. These jumps are well defined for φ in $H^1(\Omega_h)$, where $\Omega_h = \cup_{j=1, \dots, N} I_j$.

It is now easy to see that if we assume that $(T, M, \theta, w) \in [H^1(\Omega)]^4$, we have

$$-(w, \frac{d}{dx}v^1) + \langle w, \llbracket v^1 n \rrbracket \rangle = (\theta, v^1) - (\frac{1}{GA}T, v^1), \quad (3a)$$

$$-(\theta, \frac{d}{dx}v^2) + \langle \theta, \llbracket v^2 n \rrbracket \rangle = (\frac{1}{EI}M, v^2), \quad (3b)$$

$$-(M, \frac{d}{dx}v^3) + \langle M, \llbracket v^3 n \rrbracket \rangle = (T, v^3), \quad (3c)$$

$$-(T, \frac{d}{dx}v^4) + \langle T, \llbracket v^4 n \rrbracket \rangle = (q, v^4). \quad (3d)$$

for all $v^1, v^2, v^3, v^4 \in H^1(\Omega_h)$. This is the weak formulation we use to define the DG methods.

3 The DG Methods

The approximate solution $(T_h, M_h, \theta_h, w_h)$ given by the DG method will be sought in the finite dimensional space $V_h^{k_1} \times V_h^{k_2} \times V_h^{k_3} \times V_h^{k_4}$; here,

$$V_h^k := \{v : \Omega_h \mapsto \mathbb{R} : v|_{I_j} \in P^k(I_j), j = 1, \dots, N\},$$

where $P^k(K)$ is the set of all polynomials on K of degree not exceeding k . The approximate solution is determined by requiring that

$$-(w_h, \frac{d}{dx}v^1) + \langle \widehat{w}_h, \llbracket v^1 n \rrbracket \rangle = (\theta_h, v^1) - (\frac{1}{GA}T_h, v^1), \quad (4a)$$

$$-(\theta_h, \frac{d}{dx}v^2) + \langle \widehat{\theta}_h, \llbracket v^2 n \rrbracket \rangle = (\frac{1}{EI}M_h, v^2), \quad (4b)$$

$$-(M_h, \frac{d}{dx}v^3) + \langle \widehat{M}_h, \llbracket v^3 n \rrbracket \rangle = (T_h, v^3), \quad (4c)$$

$$-(T_h, \frac{d}{dx}v^4) + \langle \widehat{T}_h, \llbracket v^4 n \rrbracket \rangle = (q, v^4), \quad (4d)$$

hold for all $v^i \in V_h^{k_i}$ for $i = 1, 2, 3, 4$. To complete the definition of the method, we have to define the numerical traces $(\widehat{T}_h, \widehat{M}_h, \widehat{\theta}_h, \widehat{w}_h)$ at the nodes. It is through them that the interaction between the degrees of freedom of different intervals is introduced and the boundary conditions are actually imposed. Moreover, their choice is crucial as it affects both the stability and the accuracy of the method; see [3] for a detailed discussion of this issue for some other problems.

Extending to our framework what have been already done for fluid flow problems, we assume that the form of these traces is as follows. For an interior node $e \in \mathcal{E}_h^o$, we take

$$\widehat{w}_h = \{\{ w_h \}\} + C_{11} \llbracket w_h n \rrbracket + C_{12} \llbracket \theta_h n \rrbracket + C_{13} \llbracket M_h n \rrbracket + C_{14} \llbracket T_h n \rrbracket, \quad (5a)$$

$$\widehat{\theta}_h = \{\{ \theta_h \}\} + C_{21} \llbracket w_h n \rrbracket + C_{22} \llbracket \theta_h n \rrbracket + C_{23} \llbracket M_h n \rrbracket + C_{24} \llbracket T_h n \rrbracket, \quad (5b)$$

$$\widehat{M}_h = \{\{ M_h \}\} + C_{31} \llbracket w_h n \rrbracket + C_{32} \llbracket \theta_h n \rrbracket + C_{33} \llbracket M_h n \rrbracket + C_{34} \llbracket T_h n \rrbracket, \quad (5c)$$

$$\widehat{T}_h = \{\{ T_h \}\} + C_{41} \llbracket w_h n \rrbracket + C_{42} \llbracket \theta_h n \rrbracket + C_{43} \llbracket M_h n \rrbracket + C_{44} \llbracket T_h n \rrbracket, \quad (5d)$$

where $\{\{\varphi\}\}(e) = \frac{1}{2}(\varphi(e^+) + \varphi(e^-))$. At $x = 0$, we take

$$\widehat{w}_h(0) = w_0, \quad (6a)$$

$$\widehat{\theta}_h(0) = \theta_h(0^+) + C_{21}(0)(w_0 - w_h(0^+)) + C_{23}(0)(M_0 - M_h(0^+)), \quad (6b)$$

$$\widehat{M}_h(0) = M_0, \quad (6c)$$

$$\widehat{T}_h(0) = T_h(0^+) + C_{41}(0)(w_0 - w_h(0^+)) + C_{43}(0)(M_0 - M_h(0^+)). \quad (6d)$$

and at $x = L$,

$$\widehat{w}_h(L) = w_L, \quad (7a)$$

$$\widehat{\theta}_h(L) = \theta_h(L^-) + C_{21}(L)(w_h(L^-) - w_L) + C_{23}(L)(M_h(L^-) - M_L), \quad (7b)$$

$$\widehat{M}_h(L) = M_L, \quad (7c)$$

$$\widehat{T}_h(L) = T_h(L^-) + C_{41}(L)(w_h(L^-) - w_L) + C_{43}(L)(M_h(L^-) - M_L). \quad (7d)$$

Note how the boundary conditions are incorporated into the DG method through the definition of the numerical traces at the border. Note also that the parameters C_{ij} defining the numerical traces can have different values at different nodes. In the next section, we investigate the role of these parameters. In particular, we show that out of these sixteen parameters, six can be expressed in terms of the remaining ten and that only four of them have an impact on the “energy” of the discretization.

4 The Discrete Energy Identity

To see this, we consider a classical energy argument. It is not difficult to see that if we take $v^1 = T$, $v^2 = -M$, $v^3 = -\theta$, and $v^4 = w$ in the equations (3), and add them, we obtain the energy identity

$$\left(\frac{1}{EI} M, M\right) + \left(\frac{1}{GA} T, T\right) = (q, w) + bc(T, M, \theta, w),$$

where

$$bc(T, M, \theta, w) = M_L \theta(L^-) - M_0 \theta(0^+) - w_L T(L^-) + w_0 T(0^+).$$

Since this identity captures an essential feature of the problem under consideration, we would like to obtain a similar energy identity for the DG method. Such an identity is obtained in the following result.

Proposition 1 (Discrete Energy Identity). *Assume that $(T_h, M_h, \theta_h, w_h)$ is a solution of the DG method given by (4), with numerical traces given by (5), (6), and (7). Moreover, assume that for all nodes $e \in \mathcal{E}_h$ we have*

$$C_{21} = C_{43}, \quad -C_{22} = C_{33}, \quad C_{24} = C_{13}, \quad C_{31} = C_{42}, \quad C_{34} = C_{12}, \quad -C_{11} = C_{44}. \quad (8)$$

Then, we have

$$\left(\frac{1}{EI} M_h, M_h\right) + \left(\frac{1}{GA} T_h, T_h\right) + \Theta_{jumps} = (q, w_h) + bc(T_h, M_h, \theta_h, w_h) + \Theta_{bc}, \quad (9)$$

Here, setting $C_{14} = C_{32} = 0$ at the boundary nodes, we have

$$\begin{aligned} \Theta_{jumps} &= \sum_{e \in \mathcal{E}_h} (C_{14} [T_h n]^2 - C_{23} [M_h n]^2 - C_{32} [\theta_h n]^2 + C_{41} [w_h n]^2) (e), \\ \Theta_{bc} &= w_0 [C_{41} w(0^+) - C_{21} M(0^+)] + M_0 [C_{43} w(0^+) - C_{23} M(0^+)] \\ &\quad + w_L [C_{41} w(L^-) - C_{21} M(L^-)] + M_L [C_{43} w(L^-) - C_{23} M(L^-)]. \end{aligned}$$

Proof. The proof of the above result follows by mimicking what was done for the continuous case, that is, by taking $v^1 = T_h$, $v^2 = -M_h$, $v^3 = -\theta_h$, and $v^4 = w_h$ in the definition of the DG method (4), adding the resulting equations, and carrying out some simple algebraic manipulations. $\square$

It is now clear that if we take

$$C_{14}, \quad -C_{23}, \quad -C_{32}, \quad C_{41} \geq 0, \quad (10)$$

then each of the terms of Θ_{jumps} can be considered to be an *energy* associated with the discontinuous nature of the discretization. Thus, the above condition ensures that the appearance of the jumps in the DG approximation is accompanied by an increase of the total energy of the system. Since this can also be thought of as being a stabilizing effect, the above parameters are called the *stabilization* parameters. None of the remaining parameters appear in the expression for the energy of the approximation, as we can see in the above result.

5 Existence and uniqueness of the DG approximation

Our main theoretical result is the following.

Theorem 1 (Existence and uniqueness of the DG approximation). Consider the DG method defined by the weak formulation (4) and the numerical traces (5), (6) and (7). Assume that the constants C_{ij} satisfy (8) and (10). Then the method has a unique solution in the following cases:

- Case 1:** $C_{41} > 0$ on $\mathcal{E}_h$, $-C_{32} > 0$ on $\mathcal{E}_h^o$, $k_2 \geq k_3 - 1$, and $k_1 \geq k_4 - 1$.
Case 2: $C_{ij} = 0$ on $\mathcal{E}_h^o$, except $C_{11} = C_{22} = -C_{33} = -C_{44} = 1/2$, $C_{41}(L) > 0$, $k_2 \geq k_3$ and $k_1 \geq k_4$.
Case 3: $k_2 \geq k_3 + 1$ and $k_1 \geq k_4 + 1$.

From the first case, we see that the stabilization parameters associated to θ_h and w_h , namely, C_{32} and C_{41} , respectively, have a stronger influence on the existence and uniqueness of the solution of the method than the ones associated to T_h and M_h , namely, C_{14} and C_{23} , respectively.

From the second and third cases, we see that when the stabilization effect of a jump is turned off (by setting the corresponding stabilization parameter equal to zero), the existence and uniqueness of the approximate solution can still be guaranteed by a suitable definition of the other parameters and/or by modifying the polynomial degree of the approximate solutions. Roughly speaking, the more stabilization parameters are equal to zero, the more the spaces for θ_h and w_h have to be in relation to the spaces of M_h and T_h , respectively.

Proof. Due to the linearity of the problem, it is enough to show that the only solution to (4) with $q = 0$, $w_0 = w_L = M_0 = M_L = 0$ is $w_h = \theta_h = M_h = T_h = 0$. In this case, (9) takes the form

$$\left(\frac{1}{EI}M_h, M_h\right) + \left(\frac{1}{GA}T_h, T_h\right) + \Theta_{jumps} = 0. \quad (11)$$

By the assumptions on the parameters C_{ij} , this implies $T_h = 0$, $M_h = 0$. On the other hand, taking $v^1 = 1$ in (4a), we get that $(\theta_h, 1) = 0$.

Case 1. In the first case, from the discrete energy identity (11) we see that $[\![\theta_h n]\!] = 0$ on $\mathcal{E}_h^o$, and $[\![w_h n]\!] = 0$ on $\mathcal{E}_h$. As a consequence, if $k_3 = 0$, θ_h is a constant and since $(\theta_h, 1) = 0$, it is equal to zero. If $k_3 > 0$, a similar conclusion can be reached as follows: We have, by (5b), that $\hat{\theta}_h = \theta_h$ and, by (4b), that $(\frac{d}{dx}\theta_h, v^2) = 0$ for all $v^2 \in V_h^{k_2}$. Since $\theta_h \in V_h^{k_3}$ and $k_2 \geq k_3 - 1$, this implies that θ_h is a constant and since $(\theta_h, 1) = 0$, that it is equal to zero. A similar argument shows that w_h is also zero if $k_1 \geq k_4 - 1$.

Case 2. In the second case, by the definition of the numerical trace of θ_h , we have that $\hat{\theta}_h(0) = \theta_h(0^+)$ and that $\hat{\theta}_h(e) = \theta_h(e^-)$ for all the remaining nodes e . Taking v^2 with support (x_0, x_1) we get, by equation (4b), that $-\int_{x_0}^{x_1} \theta_h \frac{d}{dx} v^2 + (\theta_h v^2)(x_1^-) - (\theta_h v^2)(x_0^+) = \int_{x_0}^{x_1} v^2 \frac{d}{dx} \theta_h = 0$. Since $\theta_h \in V_h^{k_3}$ and $k_2 \geq k_3$ this implies that $\theta_h = 0$ on that interval. Now, if we take v^2 with support on (x_1, x_2) , equation (4b) becomes $\int_{x_1}^{x_2} \theta_h \frac{d}{dx} v^2 = \theta_h(x_2^-) v^2(x_2^-)$. Taking $v^2 = 1$, we get that $\theta_h(x_2^-) = 0$. Also, since $k_2 \geq k_3$, we can take

$v^2 = \theta_h$ and obtain that $\theta_h(x_1^+) = 0$. Hence, $\int_{x_0}^{x_1} v^2 \frac{d}{dx} \theta_h = 0$, and so $\theta_h = 0$ in (x_1, x_2) . By repeating this argument, we obtain that $\theta_h = 0$. Similarly, it is easy to show that $w_h = 0$ outside the last interval (x_{N-1}, x_N) . By the equation (4a), we have $\int_{x_{N-1}}^{x_N} v^1 \frac{d}{dx} w_h = 0$, and hence w_h is a constant on the last interval since $w_h \in V_h^{k_4}$ and $k_1 \geq k_4$. Finally, since $C_{41}(L) > 0$, by the discrete energy identity (11), we have that $[[w_h n]](L) = 0$ and so $w_h(L^-) = 0$. This implies that $w_h = 0$.

Case 3. Finally, consider the third case. Since $k_2 \geq 1$, taking $v^2 = x/L$ in (4b), we get that $\hat{\theta}_h(L) = (\theta_h, 1) = 0$. Then, taking $v^2 = 1$ on (x_j, x_{j+1}) and $v^2 = 0$ for the rest of the domain, the equation (4b) yields $\hat{\theta}_h(x_j) = \hat{\theta}_h(x_{j+1})$. This implies that $\hat{\theta}_h = 0$ on all the nodes. By (4b), this implies that $(\theta_h, \frac{d}{dx} v^2) = 0$ for all $v^2 \in V_h^{k_2}$, and since $\theta_h \in V_h^{k_3}$ and $k_2 \geq k_3 + 1$, this implies that $\theta_h = 0$. A similar argument shows that $w_h = 0$ if $k_1 \geq k_4 + 1$.

This completes the proof. $\square$

6 Numerical Results

In this section, we display preliminary numerical experiments showing that DG methods for the Timoshenko beam can be constructed so as to achieve optimal rates of convergence as well as exponential convergence. We consider two test problems. In both problems, same-degree polynomial approximations are used for all the unknowns, that is, we take $k_1 = k_2 = k_3 = k_4 = k$.

Test problem 1. We solve (1)-(2) in $\Omega = (0, 1)$ with the boundary conditions $w_0 = w_L = 0$ and $M_0 = M_L = 0$. We take $q(x) = \sin(\pi x)$, for all $x \in \Omega$. The transverse cross-section area of the beam at the point x , $A(x)$, is assumed to be rectangular with width and height $a(x)$ and $b(x)$; the moment of inertia is given by the formula $I(x) = a(x)b^3(x)/12$. We take $a(x) = b(x) = 0.01$. The Young modulus is $E = 10^7$ whereas the shear modulus is $G = (0.35)E$.

We pick the numerical traces by setting $C_{ij}(x) = 0$ for $i \neq j$ except $-C_{23}(0) = C_{41}(L) = 1/h$. And $C_{11}(x) = C_{22}(x) = -C_{33}(x) = -C_{44}(x) = 0.5$. The existence and uniqueness of the approximate solution is guaranteed by the second case of Theorem 1.

Test problem 2. The purpose of this test problem is to show that the method can easily handle other boundary conditions and discontinuities in the material properties and the load. Thus, we take the boundary conditions $w_0 = 0$, $\theta_0 = 0$, $M_L = 0$ and $T_L = 0$. We also take $q(x) = \sin(\pi x)$ if $0 \leq x \leq 1/2$ and $q(x) = -16(x - 3/4)^2 + 1$ if $1/2 < x \leq 1$. $a(x) = b(x) = 0.02$ if $0 \leq x \leq 1/2$ and 0.05 if $1/2 < x \leq 1$.

We use the following numerical traces. To capture the boundary conditions, we take $\hat{w}_h(0) = 0$, $\hat{\theta}_h(0) = 0$, $\hat{M}_h(L) = 0$, and $\hat{T}_h(L) = 0$. To define the remaining numerical traces, we simply take $C_{ij} = 0$ except for $C_{11} = C_{22} = -C_{33} = -C_{44} = 1/2$. The existence and uniqueness of the approximate solution is *not* guaranteed by Theorem 1, but can be easily proven.

in Fig. 1, we see that in both test problems, exponential convergence is achieved. In Figs. 2, and 3, the solid line represents the exact solution and “+” represents the numerical trace of the approximate solution at the nodes of the mesh. We display the results obtained on a uniform mesh of 10 elements. In the Tables 1 and 2, we display convergence orders up to polynomial degree $k = 5$. Mesh number i means a uniform mesh of 2^i elements. We can see that optimal orders of convergence are achieved for all the unknowns even in the case of piecewise constant approximation.

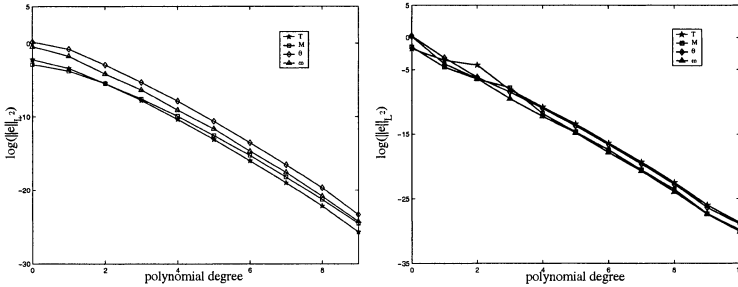


Fig. 1. Exponential convergence: Test problem 1 (left) and 2 (right)

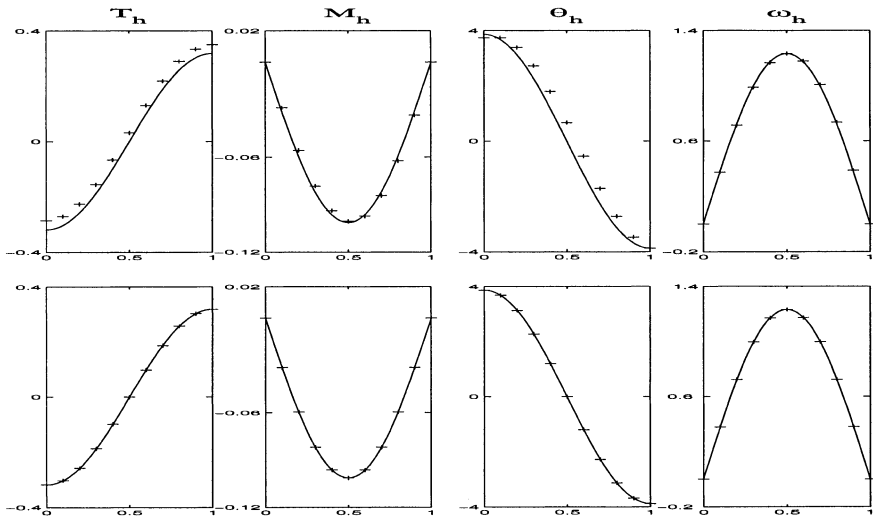


Fig. 2. Piecewise constant (top) and linear (bottom) approximations for test problem 1

Table 1. Convergence rates for same-degree approximation for test problem 1.

degree	mesh	$\ e_T\ _{L^2(\Omega)}$		$\ e_M\ _{L^2(\Omega)}$		$\ e_\theta\ _{L^2(\Omega)}$		$\ e_w\ _{L^2(\Omega)}$	
k	number	error	order	error	order	error	order	error	order
0	4	1.6e-02	0.99	9.3e-03	0.98	1.6e-01	1.00	9.5e-02	0.99
	5	8.0e-03	1.00	4.7e-03	1.00	7.9e-02	1.00	4.8e-02	1.00
	6	4.0e-03	1.00	2.3e-03	1.00	3.9e-02	1.00	2.4e-02	1.00
1	4	5.3e-04	2.00	2.5e-04	2.70	6.4e-03	1.99	2.0e-03	2.01
	5	1.3e-04	2.00	4.6e-05	2.47	1.6e-03	1.99	5.1e-04	2.00
	6	3.3e-05	2.00	1.1e-05	2.11	4.0e-04	2.00	1.3e-04	2.00
2	4	8.3e-06	3.00	2.7e-06	3.24	1.0e-04	2.99	3.2e-05	3.00
	5	1.0e-06	3.00	3.3e-07	3.04	1.3e-05	3.00	4.0e-06	3.00
	6	1.3e-07	3.00	4.1e-08	3.00	1.6e-06	3.00	5.0e-07	3.00
3	4	1.0e-07	4.00	4.9e-08	4.92	1.2e-06	3.97	3.9e-07	4.00
	5	6.3e-09	4.00	2.2e-09	4.49	7.6e-08	3.99	2.4e-08	4.00
	6	3.9e-10	4.00	1.3e-10	4.11	4.8e-09	4.00	1.5e-09	4.00
4	4	9.8e-10	5.00	3.2e-10	5.31	1.2e-08	5.00	3.8e-09	5.00
	5	3.1e-11	5.00	9.7e-12	5.04	3.7e-10	5.00	1.2e-10	5.00
	6	9.5e-13	5.00	3.0e-13	5.00	1.2e-11	5.00	3.7e-12	5.00
5	3	5.1e-10	5.99	4.9e-10	6.83	6.1e-09	5.93	2.0e-09	6.01
	4	7.9e-12	6.00	3.9e-12	6.99	9.6e-11	5.98	3.1e-11	6.00
	5	1.2e-13	5.99	4.3e-14	6.50	1.5e-12	6.00	4.8e-13	6.00

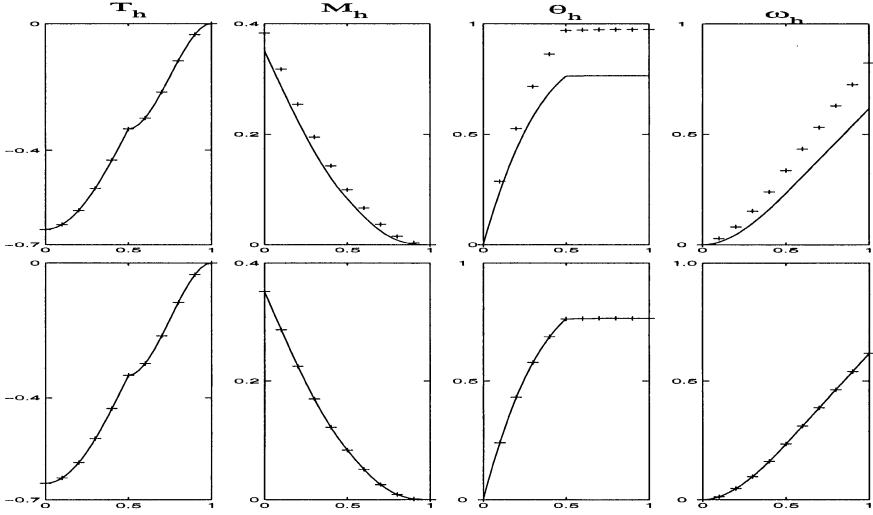

Fig. 3. Piecewise constant (top) and linear (bottom) approximations for test problem 2

Table 2. Convergence rates for same-degree approximation for test problem 2.

degree k	mesh number	$\ e_T\ _{L^2(\Omega)}$		$\ e_M\ _{L^2(\Omega)}$		$\ e_\theta\ _{L^2(\Omega)}$		$\ e_w\ _{L^2(\Omega)}$	
		error	order	error	order	error	order	error	order
0	2	9.7e-2	0.89	1.1e-1	1.04	5.7e-1	1.14	4.7e-1	1.32
	3	5.1e-2	0.94	5.6e-2	1.04	2.7e-1	1.09	2.1e-1	1.19
	4	2.6e-2	0.98	2.7e-2	1.03	1.3e-1	1.05	9.6e-2	1.10
1	2	1.3e-2	1.25	2.7e-3	1.94	1.1e-2	1.99	4.2e-3	2.04
	3	3.4e-3	1.89	6.7e-4	1.99	2.8e-3	2.00	1.1e-3	1.98
	4	8.6e-4	1.98	1.7e-4	1.99	6.9e-4	2.00	2.7e-4	1.98
2	2	1.8e-3	3.00	2.6e-4	2.61	2.8e-4	2.92	2.1e-4	2.95
	3	2.2e-4	3.00	3.4e-5	2.93	3.5e-5	2.97	2.7e-5	2.98
	4	2.8e-5	3.00	4.3e-6	2.99	4.5e-6	2.99	3.4e-6	2.99
3	2	1.8e-5	4.01	2.7e-5	3.99	1.4e-5	3.96	4.5e-6	4.06
	3	1.1e-6	4.01	1.7e-6	4.00	8.6e-7	3.99	2.8e-7	4.02
	4	7.1e-8	4.00	1.1e-7	4.00	5.4e-8	4.00	1.7e-8	4.01
4	2	6.9e-7	4.94	2.3e-7	5.05	5.3e-7	4.96	1.7e-7	4.89
	3	2.2e-8	4.98	7.1e-9	5.02	1.7e-8	4.99	5.3e-9	4.97
	4	6.9e-10	4.99	2.2e-10	5.01	5.3e-10	5.00	1.7e-10	4.99
5	2	2.3e-8	6.01	7.1e-9	5.91	1.7e-8	5.97	5.7e-9	6.04
	3	3.6e-10	6.00	1.1e-10	5.98	2.7e-10	5.99	8.9e-11	6.01
	4	5.6e-12	6.00	1.8e-12	5.99	4.3e-12	6.00	1.4e-12	6.01

7 Conclusion

We devised a family of DG methods for the Timoshenko beam problem and provided sufficient conditions for the existence and uniqueness of its solution. We then displayed numerical results showing that the method converges to the exact solution with optimal order even in the case of discontinuities of the material and the load were presented. We also provided numerical results indicating exponential convergence. In a forthcoming paper, we provide a complete error analysis of the methods and show that they are free from shear locking.

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